

Creative Hair Style Generation and Editing

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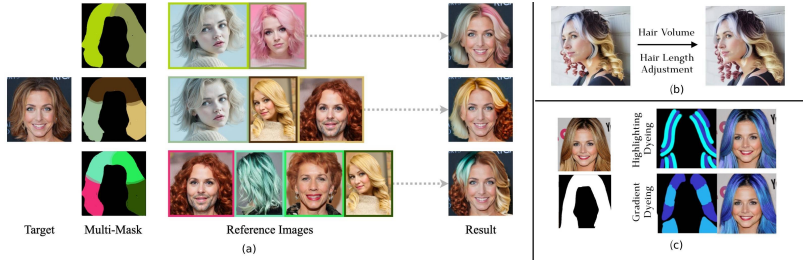


Figure 1: Main components of our method: (a) Multi-hairstyle compositing. (b) Interactive hair volume and hair length adjustment. (c) Hair color customization. All shown images are generated by the authors.

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1 Introduction

Rapid and intuitive visualization of hairstyle ideas has a substantial impact in virtual hairstyling, digital fashion, and the beauty industry, where creativity and personalization are central. Current commercial tools and academic methods often suffer from limited controllability, low diversity, or constrained editing workflows. For instance, most existing systems focus on style replacement only, providing little room for users to explore hybrid styles, personalized adjustments, or experimental designs.

In this work, we introduce a diffusion-based framework for creative hairstyle generation and editing that emphasizes both photorealism and user-driven control. Starting from a single headshot, our system allows users to compose arbitrary hairstyle references—for example, blending bangs from one exemplar with curls from another—to form novel and customizable looks. Beyond structural style transfer, we integrate stroke-based interactive color editing, enabling flexible recoloring that simulates realistic effects such as highlight dyeing, balayage, or gradient shading. This design allows exploration of both natural tones and more imaginative palettes. Moreover, an important dimension of hairstyling lies in

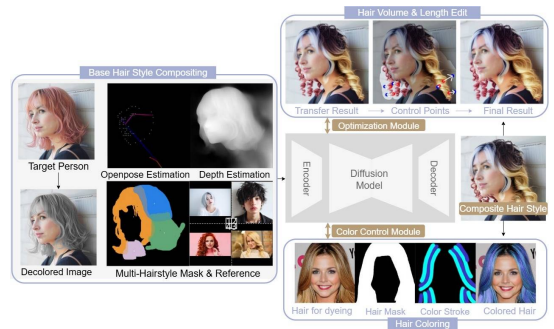


Figure 2: System overview.

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volumetric and geometric manipulation. To this end, we incorporate drag-based spatial control that empowers users to dynamically adjust hair volume and length while immediately visualizing the outcome. Such fine-grained geometric editing is crucial not only for aesthetics but also for practical applications in virtual try-on, 3D avatar design, and creative concept art. Taken together, our system provides a unified and highly configurable solution for hairstyle design. It bridges the gap between photorealistic diffusion models and intuitive, creative editing interfaces, offering a novel step toward interactive hair design tools.

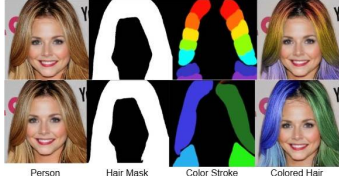


Figure 3: Hair color tone editing.

2 Method

An overview of our framework is shown in Figure 2. Our approach builds on a pre-trained diffusion model and introduces three interactive modules to facilitate diverse and customizable hairstyle design: (i) Multi-Hair Style Compositing, (ii) Stroke-based Color Tone Editing, and (iii) Drag-based Hair Volume and Length Control. These modules are designed to reflect the inherently interactive nature of hair design, enabling users to compose, recolor, and reshape hairstyles with both flexibility and realism.

Multi-Hair Style Compositing. To enable creative hairstyle composition, we introduce a geometry- and mask-guided conditional generation framework. Pose and depth maps guide the geometry, ensuring transferred styles align with head orientation and 3D structure. Each style exemplar is encoded (e.g., via IP-Adapter [Ye et al. 2023]) into hairstyle embeddings, which are blended in latent diffusion through cross-attention over overlapping regions, modulated by user-defined parameters. As shown in Figure 1(a), this enables intuitive combination of multiple references (e.g., bangs from one and curls from another) into a coherent hairstyle.

Stroke-based Color Tone Editing. For hair color editing, in Figure 3, we adopt the idea of stroke-based guidance from ControlColor [Liang et al. 2024]. Users simply draw a few rough strokes on the hair to indicate the desired colors. These strokes are then converted into color hints with masks that tell the model exactly where to apply them. During the diffusion process, the model aligns the strokes with the hair structure so that the colors blend naturally with the strands, rather than spilling into the face or background. This allows users to try out a wide range of looks, from subtle highlights to bold gradient dye effects, while keeping the results realistic and faithful to the original hairstyle.

Drag-based Hair Volume and Hair Length Control. To manipulate hair geometry, we extend DragDiffusion [Shi et al. 2023], where users specify source points and target points. The diffusion model updates the generation such that features near source move toward

target, allowing interactive adjustment of hair length and volume. To prevent distortion of facial identity, we introduce fixed facial control points (e.g., eyes, nose, mouth corners) with identity mapping constraints, anchoring the facial region during manipulation. We observe that this is the key to maintaining facial identity.

3 Results and Conclusion

Figure 1 (a) shows the result of Multi-Hair Style Compositing. The target was transformed using two, three, and four different hairstyles. These results comprehensively demonstrate the efficacy of our approach across a diverse range of hairstyle transfer tasks. The outer frame of the reference images and the corresponding position of the mask are the same color, which is used to represent the hairstyle filled in each position.



Figure 4: Additional hair volume and hair length editing results. (a) Hair volume increase. (b) Hair shortened.

Figure 3 presents the results of stroke-based colorization. Users are able to control hair color in a fine-grained and location-specific manner. The injected colors blend naturally into the hair while preserving texture details, yielding photorealistic results without noticeable color bleeding or artifacts. Figure 4 shows examples of volume and length editing. With simple drag commands, hair volume can be increased or reduced, and hair length can be shortened while preserving the natural shape of the head and facial features. In summary, our framework provides a unified diffusion-based solution for comprehensive, controllable, and photorealistic hairstyle editing, seamlessly integrating structural compositing, color customization, and geometric manipulation. This flexibility opens new opportunities for virtual hairstyling, fashion design, and creative digital applications. While our current work focuses primarily on hair, performance may degrade when large regions of background or clothing are included in the edits. Addressing these broader editing scenarios will be an important direction for future work.

Acknowledgments

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